

Analyzing Clinical Sentences to Determine and Prevent Bias in AI Training Algorithm

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1 Introduction

AI usage has increased dramatically over the last half a decade. Although the use of AI has tremendous benefit such as quick and easy concise summary, easier access to curated data, and generation of novel unstructured data (images, videos and text). The efficacy of AI use is as important as its training. Since AI are trained on vast amount of corpus data. The training regimen is crucial in understanding and eliminating bias. The AI cycle can be roughly divided into conception, data collection, pre-processing, in-processing, post-processing and deployment. There are various types of biases found in every step across the AI mode life cycle. For instance during conception, biases include implicit bias, sensitive attribute bias, confirmation bias, systemic bias. Biases found during the data collection includes representation bias, sampling bias and selection bias. Other biases found in the pre-processing cycle stage include: feature selection bias, representation bias, aggregation bias. In-processing biases includes algorithmic bias and validation bias, post-processing biases includes evaluation bias and predictive bias, feedback loop bias and dismissal bias can be seen after the AI model is deployed

Our project focuses on implicit biases which is a conceptual biases seen during the initial training of AI model. We attempt to analyze clinical sentences showing various types of bias among a different subset of categories. We performed few classification algorithm and sentimental analysis to give us insight into bias among each sentences. We then extrapolated those bias and discuss useful modification to prevent bias in a clinical setting, the ethics, potential harm and privacy of AI usage in the medical field.

1.1 Problem Statement

The specific problem our project addresses is ways to reduce implicit bias, when training AI model, the ethics issue such as privacy that arise with AI usage in health care and the potential harm and ways to mitigate them.

1.2 Research Question

The goal of our project is to investigate how biases during clinical assessments lead to AI bias? AI are trained on datasets that often constitute some form of human bias during the initial conception. Our goal is to analyze this bias and find ways to alleviate or mitigate bias in machine learning training?

1.3 Contributions

Our project contributes the following:

- We performed some data inspection by cleaning duplicate columns before doing some Exploratory Data Analysis to look for insight in the datasets
- We compare four classification models on a bias clinical dataset (Logistic Regression, KNN classifier, Random Forest classifier and Gradient Boosting Classifier).
- We apply a full pre-processing pipeline, including cleaning, tokenization, n-grams, using CountVecotrizer via Natural Language Processing .
- We performed sentimental analysis to check for emotional context associated with the type of bias.
- We discussed way to prevent bias by recommending fairness metrics and other algorithms that use diversified dataset to prevent sampling bias when training AI models
- We highlight the ethics, privacy, and harm associated with research when training AI models.

1.4 Paper Structure

The rest of the paper is organized as follows: Section 2, shows the EDA performed. Section 3, describes our methodology, including the dataset, preprocessing steps, and model setup. Section 4, presents the quantitative results, Section 5, discusses the qualitative results and, Section 6 presents the summary and conclusion. Finally, Section 7

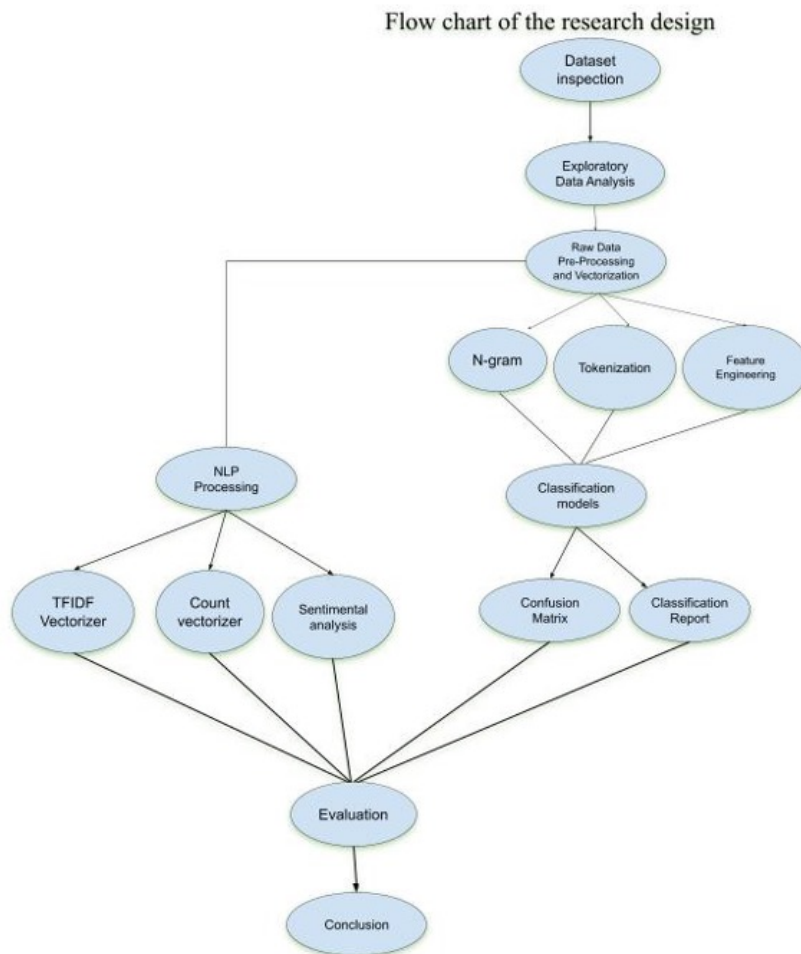


Figure 1: A flow chart highlighting the design of the paper

2 Exploratory Data Analysis

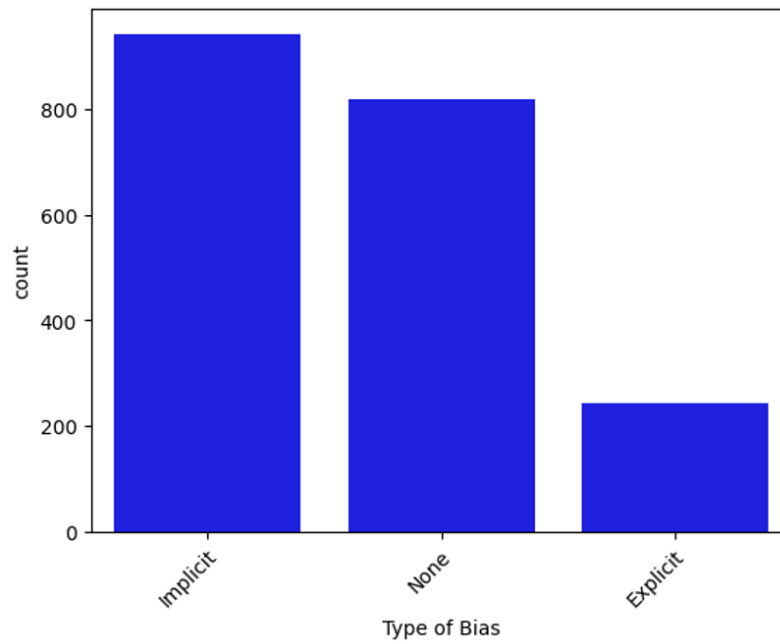


Figure 2: Count plot showing the various types of bias.

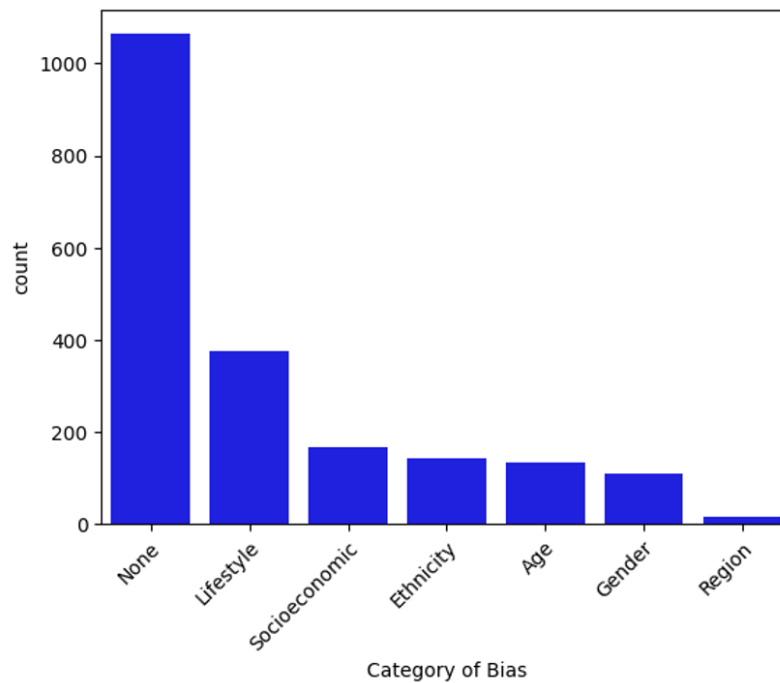


Figure 3: Count plot showing the various categories of bias.

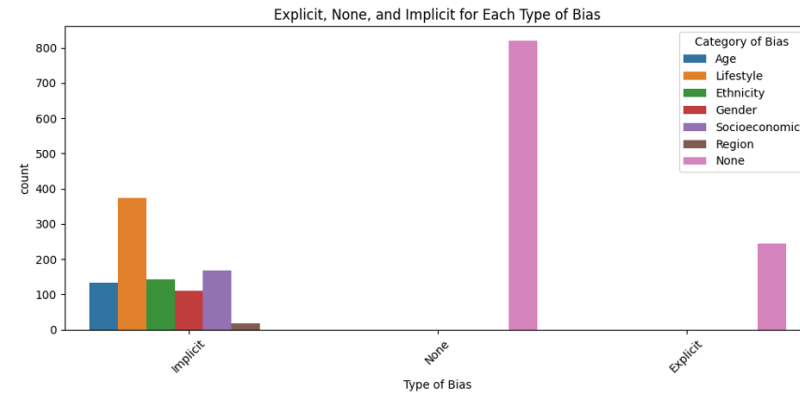


Figure 4: Count plot showing explicit, none, and implicit bias for each category.

3 Methodology and Approach

The design of the research is broken into various section. First, we did some EDA to provide insight into the data after inspection, we then performed preprocessing using vectorization which handles n-gram, tokenization and feature engineering before performing predictive modelling using a classification analysis on 4 models (logistic regression, KNeighbor Classifier, RandomForest Classifier and GradientBoosting Classifier. An evaluation metrics was conducted to predict if the target variable (category of bias) influence types of bias. A confusion matrix and a F1-score was derived.

Finally, NLP was used to analyze the text (sentences) using Countvectorizer and TFIDVectorizer to analyze the frequency of word and implicit discriminative words leading to biases. A sentiment analysis was conducted which analyze the sentence based on categories and scored them which determine the sentiment_label (positive, negative or neutral).

3.1 Dataset Information

The dataset was derived from Kaggle as a csv file. It constitutes of three object columns divided into **sentences**, **type of bias** (implicit, explicit or none) and **category of bias** (None, lifestyle, socioeconomic, ethnicity, age, gender and region). The dataset consists of 2007 clinical phrases and entries.

3.2 Preprocessing Steps

Data inspection, there were some redundant columns present in the dataset that had to be combined to remove duplicate columns.

Pre-processing was done using NLP vectorization that automatically does tokenization, feature engineering and N-gram.

3.3 Classification Models Used

Four different models were applied:

3.3.1 Logistic Regression (LR)

3.3.2 KNeighbor Classifier (KNN)

- Hyperparameter 1: n_neighbors=5, weight=uniform,
- Hyperparameter 2: n_neighbors=10, weight=uniform,

- Hyperparameter 3: `n_neighbors=25, weight=uniform`

3.3.3 RandomForest Classifier (RF)

- Hyperparameter 1: `n_estimator=100, max_leaf_node=30,`
- Hyperparameter 2: `n_estimator=100, max_leaf_node=50,`
- Hyperparameter 3: `n_estimator=100, max_leaf_node=70`

3.3.4 GradientBoosting Classifier (GB)

- Hyperparameter 1: `n_estimators=100, learning_rate=0.1, max_leaf_nodes=3,`
- Hyperparameter 2: `n_estimators=100, learning_rate=0.1, max_leaf_nodes=10`

4 Quantitative Discussion & Results

4.1 AGE

The result from the classification analysis indicates that Gradient boost using the first hyperparameter and logistic regression had the highest F1-score, precision and recall when measuring the categorical bias (AGE). It identify the age bias accurately with few false positive and few false negative in the sentences found in the bias dataset text. Sentimental analysis analyzed the emotional tone of each text associated with age to determine if it is positive, negative or neutral by computing a senntiment score. Thee sentiment label indicate that of all the sentences analyzed that had implicit bias, the vast majority were positive, while a few were both negative and neutral.

4.1.1 Table 1: Classification report evaluation metrics (Precision, Recall, and F1-score) for AGE.

Algorithm	Parameter	Precision	Recall	F1
Logistic Regression	*	0.92	0.89	0.91
KNN	1	1.00	0.30	0.46
	2	1.00	0.15	0.26
	3	1.00	0.11	0.20
Random Forest	1	0.95	0.74	0.83
	2	0.95	0.74	0.83
	3	0.95	0.78	0.86
Gradient Boosting	1	0.89	0.93	0.91
	2	0.86	0.93	0.89

4.1.2 NLP Processing: Age Graphical Representation

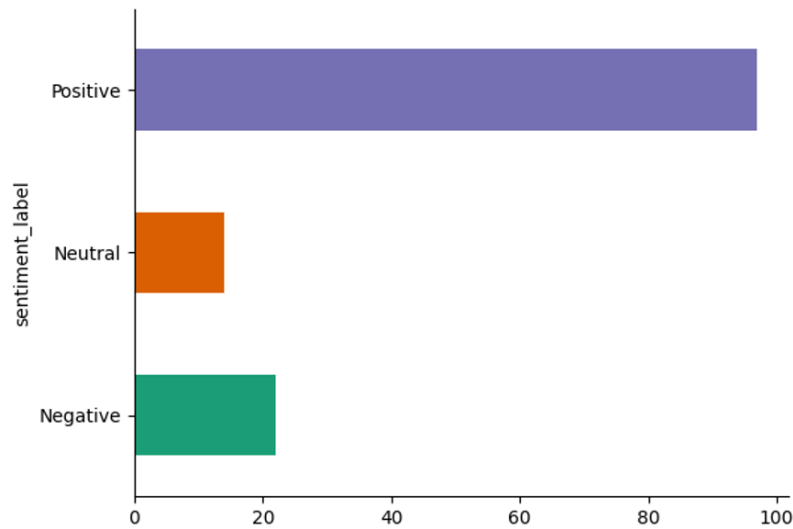


Figure 5: Sentiment label showing the frequency of type of bias across age

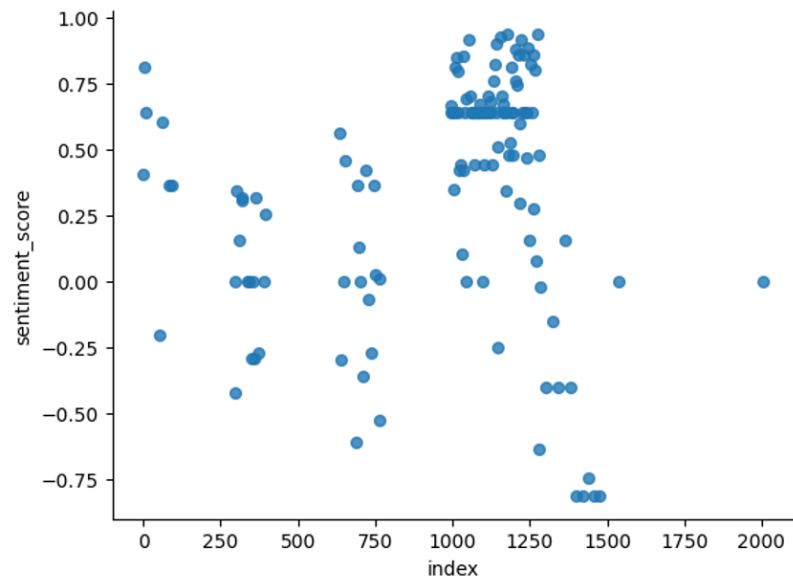


Figure 6: Sentiment score across all age

4.2 ETHNICITY

Similar to the result seen in the age categorical bias, the classification analysis for the ETHNICITY report indicates that Gradient boost using the first hyperparameter and logistic regression had the highest F1-score, precision and recall. KNN had the poorest performance. Interpretation of the result suggest the classification algorithms predicts accurately with few false positive and few false negative the sentences found in the bias dataset text. Sentimental analysis indicates that most implicit bias were positive, followed by a lesser amount of neutral statements and few negative bias statements.

4.2.1 Table 2: Classification report evaluation metrics (Precision, Recall, and F1-score) for ETHNICITY.

Algorithm	Model	Precision	Recall	F1
Logistic Regression	*	0.91	0.71	0.80
KNN	1	0.89	0.29	0.43
	2	1.0	0.21	0.35
	3	1.0	0.07	0.13
Random Forest	1	0.94	0.57	0.71
	2	0.94	0.57	0.71
	3	0.89	0.61	0.72
Gradient Boosting	1	0.90	0.68	0.78
	2	0.90	0.64	0.75

4.2.2 NLP Processing: Ethnicity Graphical Representation

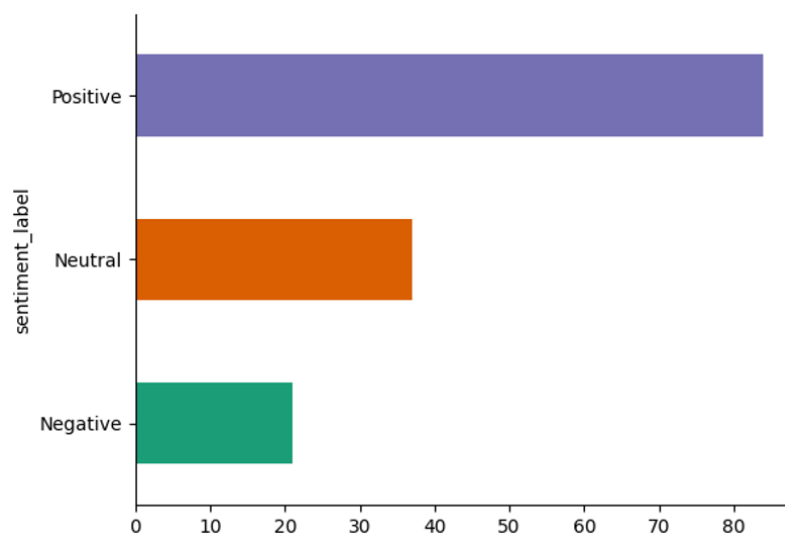


Figure 7: Sentiment label showing the frequency of type of bias across ethnicity

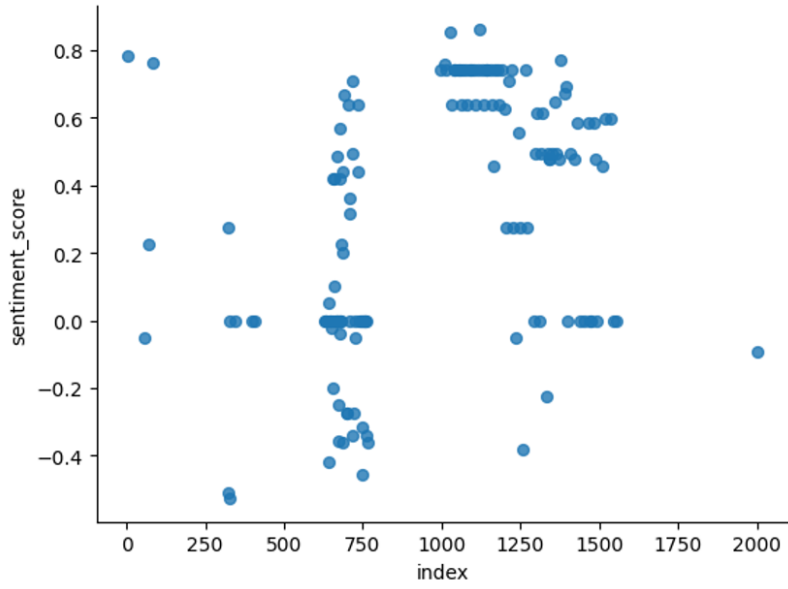


Figure 8: Sentiment score across all ethnicity

4.3 GENDER

For the GENDER categorical bias, all classification algorithm had poor performance with the exception to Gradient Boosting. The report indicates that Gradient boost using the first and second hyperparameter had the highest F1-score, precision and recall. Sentimental analysis still indicates that most implicit bias were positive. However, there was a sharp increase in negative bias in the statement analyzed even outpacing as seen in the horizontal barplot.

4.3.1 Table 3: Classification report evaluation metrics (Precision, Recall, and F1-score) for GENDER.

Algorithm	Parameter	Precision	Recall	F1
Logistic Regression	*	0.62	0.59	0.60
KNN	1	0.59	0.45	0.51
	2	0.75	0.41	0.53
	3	0.78	0.32	0.45
Random Forest	1	0.75	0.27	0.40
	2	0.75	0.27	0.40
	3	0.80	0.36	0.50
Gradient Boosting	1	0.77	0.77	0.77
	2	0.74	0.77	0.76

4.3.2 NLP Processing: Gender Graphical Representation

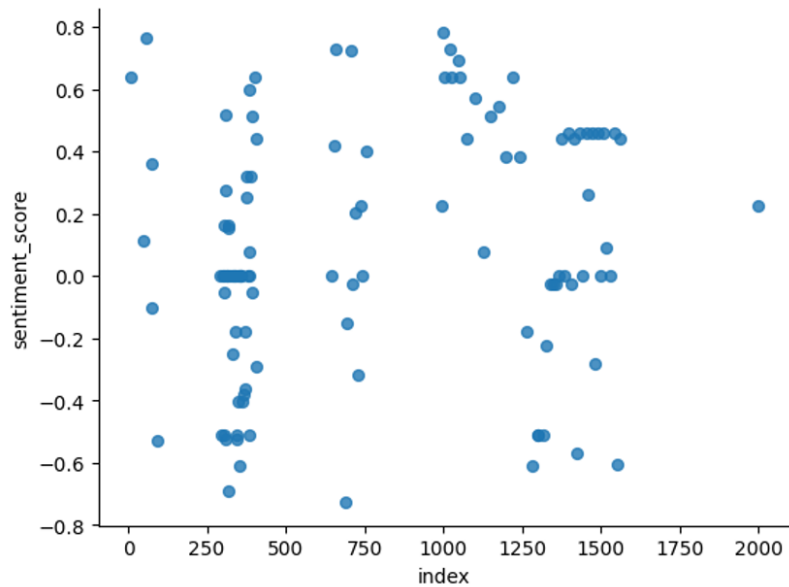


Figure 9: Sentiment score across all gender

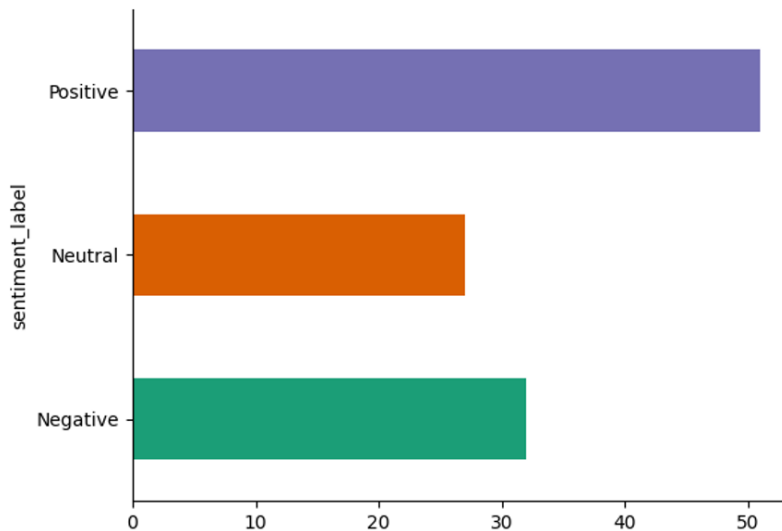


Figure 10: Sentiment label showing the frequency of type of bias across gender

4.4 LIFESTYLE

For the LIFESTYLE categorical bias, KNN classification algorithm had the worst performance while Gradient Boosting (second hyperparameter) and logistic regression had the highest precision, recall and F1-score (best performance). Sentimental analysis indicates that most implicit bias were vastly negative when it came to the emotional context of the clinical sentences analyzed. We speculate that implicit bias might affect the way clinicians addresses their patients base on their lifestyle. However, there was still a noticeable amount of positive result seen in the analysis as observed in the horizontal barplot.

4.4.1 Table 4: Classification report evaluation metrics (Precision, Recall, and F1-score) for LIFESTYLE.

Algorithm	Parameter	Precision	Recall	F1
Logistic Regression	*	0.82	0.80	0.81
KNN	1	0.84	0.48	0.61
	2	0.94	0.40	0.56
	3	0.83	0.13	0.23
Random Forest	1	0.71	0.55	0.62
	2	0.74	0.65	0.70
	3	0.77	0.72	0.74
Gradient Boosting	1	0.72	0.77	0.75
	2	0.77	0.79	0.78

4.4.2 NLP Processing: Lifestyle Graphical Representation

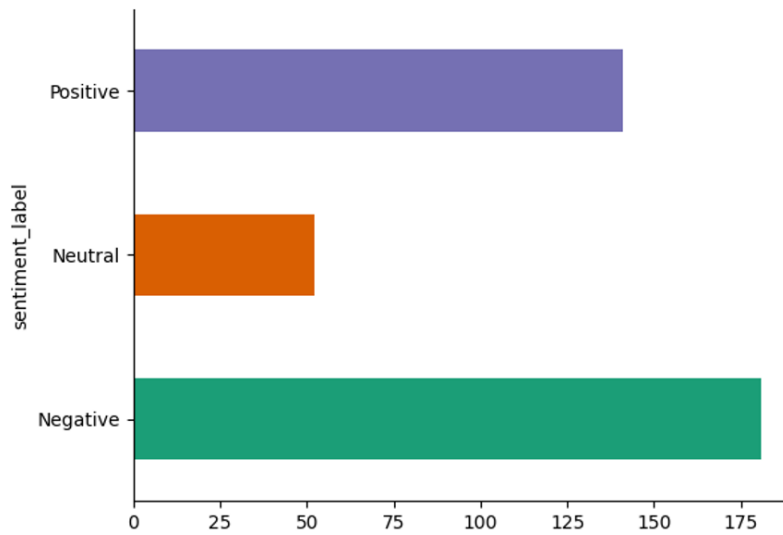


Figure 11: Sentiment label showing the frequency of type of bias based on lifestyle

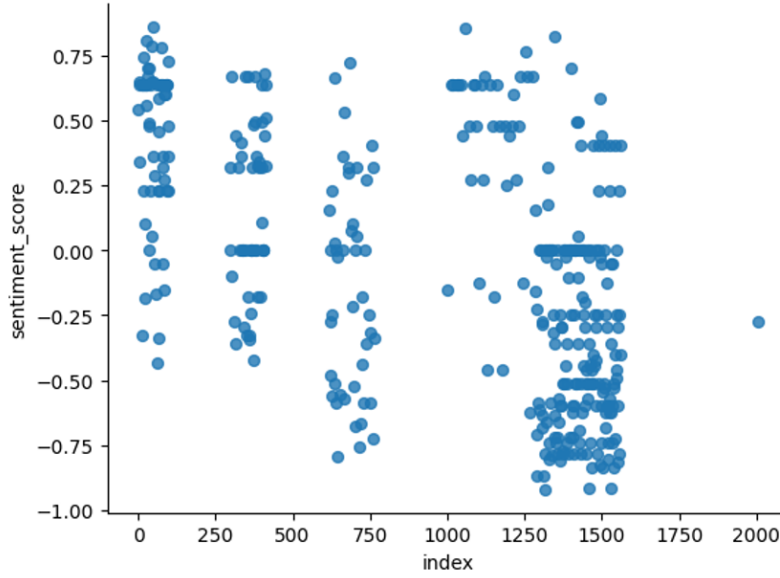


Figure 12: Sentiment score across various on lifestyle

4.5 SOCIOECONOMIC

The SOCIOECONOMIC categorical bias paints a different picture. Similar to other categorical bias, KNN classification algorithm had the worst performance while Gradient Boosting (second hyperparameter) and logistic regression had the highest precision, recall and F1-score (best performance). Sentimental analysis indicates that most implicit bias were vastly positive when it came to the emotional context of the clinical sentences analyzed. We speculate that implicit bias have a positive impact on how clinicians address their patients, the subjective reason why this might occur however is unknown.

4.5.1 Table 5: Classification report evaluation metrics (Precision, Recall, and F1-score) for SOCIOECONOMIC STATUS.

Algorithm	Parameter	Precision	Recall	F1
Logistic Regression	*	0.96	0.74	0.83
KNN	1	0.95	0.56	0.70
	2	1.00	0.35	0.52
	3	1.00	0.26	0.42
Random Forest	1	0.94	0.44	0.60
	2	0.95	0.62	0.75
	3	1.00	0.56	0.72
Gradient Boosting	1	1.00	0.74	0.85
	2	1.00	0.76	0.87

4.5.2 NLP Processing: Socio Economic Graphical Representation

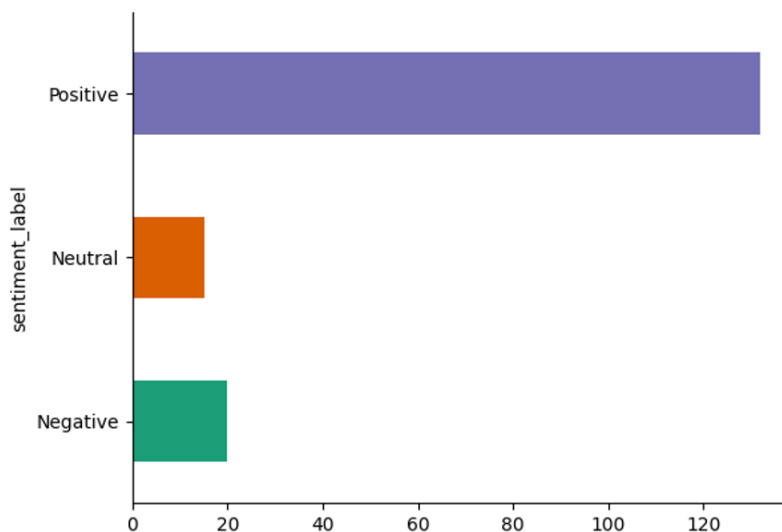


Figure 13: Sentiment label showing the frequency of type of bias based on lifestyle

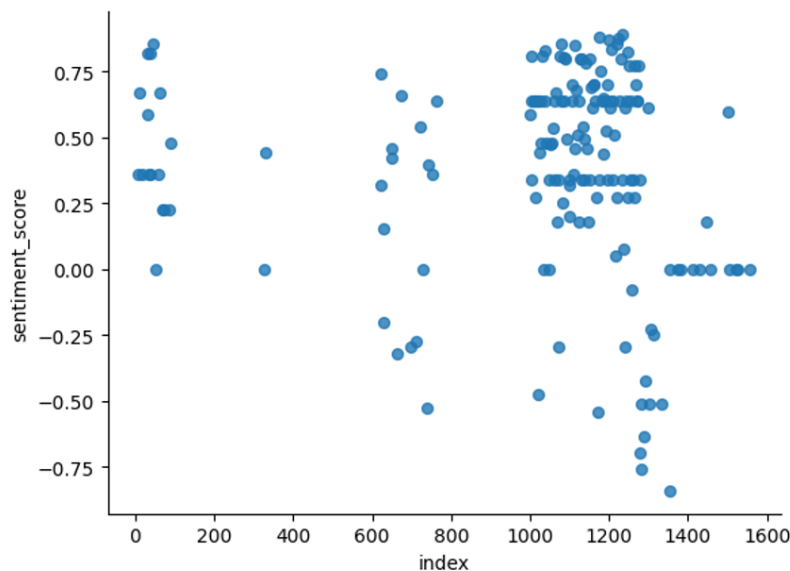


Figure 14: Sentiment score across various on lifestyle

4.6 REGION

The REGION categorical sentiment label show a positive score edging out both negative and neutral score. The classification report is no-existence. The count of the region categorical bias column shows a modest 17 association or record in the dataset. The limited amount of data regarding this categorical bias makes it difficult to run an efficient algorithm. Gradient boosting using the second hyperparameter produce a result, however the validity of this result is questionable. Sentimental analysis of the 17 clinical statements indicates that most implicit bias were vastly positive when it came to the emotional context of the clinical sentences analyzed.

4.6.1 Table 6: Classification report evaluation metrics (Precision, Recall, and F1-score) for REGION.

Algorithm	Parameter	Precision	Recall	F1
Logistic Regression	*	0.00	0.00	0.00
KNN	1	0.00	0.00	0.00
	2	0.00	0.00	0.00
	3	0.00	0.00	0.00
Random Forest	1	0.00	0.00	0.00
	2	0.00	0.00	0.00
	3	0.00	0.00	0.00
Gradient Boosting	1	0.00	0.00	0.00
	2	0.50	0.33	0.40

4.6.2 NLP Processing: Region Graphical Representation

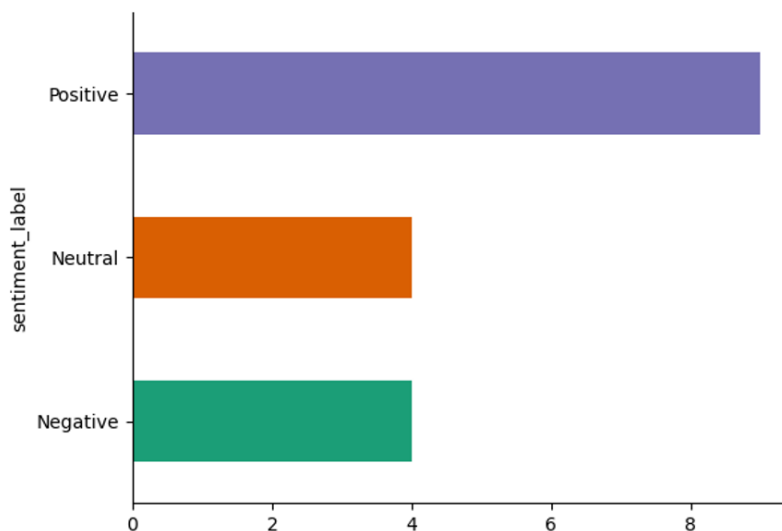


Figure 15: Sentiment label based on the region of an individual

4.7 Results Overall

After analyzing each categorical bias independently, We chose to explore the strength of each classification algorithm by plotting the weighted average of the accuracy, precision, recall and F1-score. This metrics informs us on which algorithm is well suited for analyzing clinical sentences. Not surprisingly, logistic regression and Gradient Boosting performed the best which alludes to these algorithms as useful metrics for our dataset. KNN had the poorest performance and should be interpreted with caution.

4.7.1 Table 7: Classification report evaluation metrics showing the weighted (Precision, Recall, and F1-score) for all categorical bias.

Algorithm	Accuracy	Precision	Recall	F1
Logistic Regression	0.85	0.84	0.85	0.84
KNN	0.71	0.75	0.71	0.68
	0.67	0.77	0.67	0.62
	0.60	0.72	0.60	0.51
Random Forest	0.75	0.77	0.75	0.73
	0.78	0.79	0.78	0.77
	0.80	0.81	0.80	0.79
Gradient Boosting	0.85	0.85	0.85	0.84
	0.86	0.86	0.86	0.85

4.8 Limitations and Steps to Prevent Bias in AI training

Our project encountered some limitations, most strikingly, the unstructured bias dataset was not split into various categorical labels (such as the sex of the gender, or the race of the ethnicity etc..) making it difficult to analyze bias among various demographics. For instance, our data predicted an increase in negative bias among gender. It would be useful to know if this is skewed towards a particular gender (male or female).

We propose various ways to prevent bias in training of AI model used in clinical setting. First, we recommend the use of diverse datasets from various sources when training AI model. Retrieval Augmented Reality (RAG) is preferred due to his ability to cipher large diverse datasets from various sources before creating a response.

Fairness metrics is encouraged to check for bias when training of the model. Some metrics and tool that could be used are disparate impact, demographic parity, group-wise accuracy should also be incorporated to analyze disparity among AI model based on sensitive attributes like (age, race and gender).

The table below shows some of the most common AI life cycle algorithms and the remedies to improve fairness in the AI model. Our project paper went in-depth on implicit bias which is a conceptual bias lurking around during data creation. Since implicit bias is an unconscious bias inherently unique to human. This makes it very difficult to detect. In our health care model we propose provide training to ensure implicit biases is brought into the fore-front of health care providers leading to a conscious effort to address an issue that is subtle but could have detrimental effects when it comes to treatment plans.

4.8.1 Table 8: Biases in the AI model cycle and strategies to prevent each biases.

AI life cycle	Biases	Potential Mitigation
Conception	Implicit bias	Review policies for unrecognized bias. Provide training for clinicians
	Systemic bias	Recognize the impact of historical discrimination
	Confirmation bias	Utilize multiple perspective in model design
	Sensitive attribute bias	Stay mindful of unconscious assumptions about age, birth sex, gender, ethnicity, race, sexual orientation, religion in the research question.
Data Collection	Representation bias	Collect data across diverse geographic locations, healthcare settings, and patient demographics. Actively include underrepresented groups.
	Sampling bias	Match sampling frame with target population, simplify surveys, follow up with non-responders, avoid convenience sampling, and use random sampling.
	Selection bias	Use stratified sampling for diversity. Apply blinding to prevent bias, and pre-register studies for transparency.
Pre-Processing	Aggregation bias	Utilize disaggregated data, apply regression analysis, perform sensitivity analysis, implement hierarchical modeling, and adopt alternative measures.
	Feature selection bias	Select features based on their relevance and potential impact on model outcomes, carefully avoiding assumptions based on stereotypes or incomplete information.
	Representation bias	Employ data augmentation techniques, such as generating synthetic data or using data interpolation, and apply fairness-aware data pre-processing techniques, such as re-weighting or re-sampling, to balance the dataset.
In-Processing	Algorithm bias	Conduct periodic algorithmic evaluations (Red Teaming). Proactively address and reduce biases originating from previous phases
	Validation bias	cross-validation, ensure diverse data splits, apply stratified sampling, monitor performance metrics.
Post-Processing	Evaluation bias	Employ multiple metrics to evaluate model performance and fairness. Include fairness metrics, sensitivity, specificity, and other relevant outcomes for a holistic view of model behaviour.
	Predictive bias	Post-hoc model adjustment. Use statistical techniques to adjust the model's outputs to correct for biases.
Post-Deployment	Automation bias	Educate users on critically evaluating AI recommendations. Design features that actively engage users and promote critical thinking.
	Dismissal bias	Continuously monitor and update AI predictions.
	Feedback loop bias	Provide training for healthcare professionals to understand and interact with AI systems effectively.

5 Qualitative Discussion: Ethics and Professional responsibility

5.1 Ethics

The ethical concern in this research shows how bias in clinical text can influence machine-learning models that are later used to support healthcare decision-making. The dataset used in this research contains clinical-style sentences that reflect real human judgment, it is important to recognize that the language itself can carry implicit or explicit biases related to age, gender, socioeconomic status, lifestyle, or ethnicity. Even though the dataset is publicly available and fully de-identified, we still treat it as sensitive to health-related text. The responsibility is not just to analyze the data, but to ensure we are not reproducing patterns that could harm real patients if these models were used in practice (Cross et al., 2024).

A major ethical issue is that biased inputs produce biased outputs. Research has shown that AI systems trained on imbalanced or biased clinical data can reinforce health disparities and influence diagnostic accuracy, treatment plans, and patient trust. The article *Bias in Medical AI: Implications for Clinical Decision-Making* emphasizes that biases at the data or modeling stage can lead to unequal outcomes and even worsen disparities among historically marginalized groups. The article points out that bias in medical AI can influence real clinical decisions, which is why fairness and transparency matter when models are being developed. Small issues in the data can end up affecting how patients are treated, especially groups that already deal with unequal care (Cross et al., 2024).

For this project, reviewing how different types of bias appeared in the dataset helped show where the data was uneven. Some categories had many examples, while others barely had any, and that kind of imbalance changes what a model picks up on. Noting these patterns makes it easier to understand where the model might struggle or repeat something inaccurate (Cross et al., 2024).

If a system like this was ever used in a real clinical setting, it would need ongoing monitoring. It wouldn't be safe to rely on it without people reviewing the outputs and without using more balanced and representative data. AI in healthcare can be influenced by the way information is collected and labeled, so paying attention to those limits is important. Keeping these issues in mind helps support safer and more fair use of AI tools in clinical environments (Cross et al., 2024).

5.2 Privacy

The utilization of clinical sentences and patient assessment data for artificial intelligence training presents significant privacy challenges that extend beyond traditional data protection measures. When developing machine learning models for healthcare applications, researchers must navigate the complex landscape of patient confidentiality while ensuring that datasets remain sufficiently robust for meaningful analysis [1]. The sensitive nature of clinical information requires implementing comprehensive privacy frameworks that protect individual patient identities throughout the entire research life cycle, from initial data collection to model deployment and beyond.

Patient privacy concerns become particularly acute when dealing with unstructured clinical text, as these narratives often contain detailed personal information that standard de-identification techniques may not adequately address (Alderman et al., 2025). Clinical sentences frequently include implicit identifiers such as rare medical conditions, specific treatment histories, or unique combinations of demographic and clinical characteristics that could potentially lead to patient re-identification. Furthermore, the iterative nature of machine learning model training creates multiple opportunities for privacy breaches, as datasets are repeatedly accessed, processed, and analyzed across different computational environments (Chen et al., 2024). Researchers must therefore implement robust anonymization protocols, including advanced techniques such as differential privacy and synthetic data generation, to ensure that individual patient information remains protected while preserving the statistical properties necessary for effective model training. Beyond individual privacy, the broader concern relates to how AI systems might inadvertently perpetuate surveillance-like dynamics in healthcare. For example, *The Lancet Digital Health* (2024) warns that poorly governed AI systems may normalize the extraction and secondary use of patient data without meaningful consent, potentially eroding trust between patients and clinical institutions (Alderman et al., 2025). Moreover, privacy risks in this project are amplified by the nature of the dataset itself, which contains clinical sentences labeled by bias type and demographic category—including age, gender, ethnicity, socioeconomic status, and lifestyle. As shown throughout the quantitative and NLP analyses in the Figures (5–15), these categories inherently involve sensitive attributes that could enable patient re-identification if proper safeguards are not in place. Even though the dataset used in this study was sourced from Kaggle and not drawn directly from electronic health records, research indicates that textual data containing demographic cues can still expose individuals to privacy breaches, especially when combined with external datasets or when adversarial actors apply linkage attacks (Carey et al., 2024). This reinforces the need for privacy-preserving pre-processing techniques—such as removing or masking

sensitive descriptors, implementing token-level de-identification, and validating outputs for unintended leakage—before any model development or sentiment analysis is performed.

The ethical responsibility extends beyond mere compliance with privacy regulations to encompass a broader commitment to maintaining patient trust in healthcare AI systems. Healthcare institutions and researchers must establish transparent governance frameworks that clearly define data access protocols, specify retention periods, and outline procedures for secure data disposal. Additionally, the potential for model outputs to inadvertently reveal sensitive information about training data necessitates careful evaluation of model behavior and implementation of privacy-preserving inference techniques (Rodriguez et al., 2025). By prioritizing patient privacy throughout the AI development process, researchers can contribute to building more trustworthy and ethically sound healthcare AI systems that benefit patients while respecting their fundamental right to privacy.

5.3 Potential Harm

Algorithmic bias presents several potential dangers, beginning with the risk of unequal treatment. This occurs when systems trained on incomplete or historically skewed data make decisions that favor some groups while disadvantaging others. This is dangerous because it transforms existing inequalities into automated outcomes, allowing discrimination to occur at a faster scale and often without clear visibility. When these systems affect who gets access to care, support, or resources, the harm becomes deeply embedded in everyday decision-making (Mitchell et al., 2024).

Another danger is the worsening of existing health inequities. When biased algorithms underestimate the needs of certain communities or misinterpret their health risks, those groups receive less care than they require. This is dangerous because it leads to unequal health outcomes, delayed diagnoses, and reduced access to essential services. Instead of improving fairness in health systems, poorly designed algorithms can intensify the very disparities they were intended to reduce, especially for marginalized populations (Sulaiman, 2020).

A third danger lies in the erosion of trust between patients and health care institutions. When people notice that automated systems routinely produce outcomes that feel unfair or inaccurate, confidence in those institutions declines. This is dangerous because trust is essential for effective care. Patients who do not trust the system are more likely to delay treatment, avoid screenings, or disengage from services altogether. This avoidance can lead to worsened conditions and higher long-term health risks.

Bias also carries the danger of reinforcing stigma within health care. When systems reflect prejudices such as assumptions about compliance, behavior, or worthiness of care, they influence how providers view certain patients. This is dangerous because stigma affects the quality of treatment and can cause clinicians to unconsciously offer less compassion, fewer options, or lower-quality interventions. Over time, stigmatized groups face poorer health outcomes due to discriminatory attitudes embedded in both human and automated systems (Sulaiman, 2020).

A final danger is the long-term entrenchment of structural inequality. Once biased systems are widely adopted, they continually feed biased data back into future decisions, creating a self-perpetuating cycle. This is dangerous because it becomes increasingly difficult to detect, challenge, or correct discriminatory outcomes when they appear to originate from neutral technology. The result is a system that repeatedly disadvantages the same communities while appearing fair on the surface.

6 Conclusion

Our project explored the role of bias in AI training algorithm on a clinical sentence text. Particularly, we navigated the role of implicit bias that is found during the conception of data in the AI life cycle. A classification report was used as an evaluation metrics to determine the precision, recall and F1-score in the categorical bias column of our dataset. From our classification report, gradient boosting classifier was found to be the optimal model of the 4 algorithmic models due to its performance metrics on the F1-score, precision, recall and accuracy. An analysis of the sentences was performed using natural language processing to check for the most common word and discriminative word. A sentiment analysis was performed to determine the emotional context of each word. From the sentimental analysis, lifestyle seen in the categorical bias column was shown to

have the highest amount of negative score. Socioeconomic status saw the highest amount of positive score. The limitation we stumbled upon relied on the lack of sensitive attribute which prevented us from conducting a fairness metrics.

A thorough investigation of the ethics, privacy and potential harm of biases in AI algorithm was explored, The dominant ethical issue that relates to implicit bias in our dataset is the marginalization of historical vulnerable individuals leading to disparity in the their treatment.

Privacy with regards to the use of AI was also investigated. The major takeaway message relates to the use of private sensitive clinical medical information to train AI data which can reveal implicit identifiers like rare medical conditions, treatment history, demographic traits such as sexual orientation, all which can further lead to further stigmatization.

Finally, the potential harm with the usage of AI was discussed, regarding to the paper topic; implicit bias can cause a worsen of health inequities, delay diagnoses and reduce access to essential services. Other reparable harm that can be caused could be disenfranchisement with the health system, delay treatment, screening avoidance and lack of trust directed at medical providers due to prejudicial biases.

To summarize, the paper highlights the bias that can lead to severe negative outcomes if not mitigated and offers a solution to raise awareness in order to improve the quality of health care as it transitions into the AI age.

7 References

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